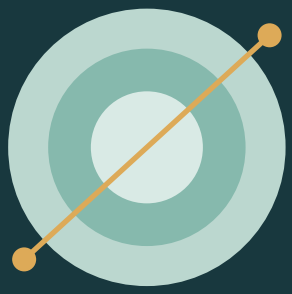
Lumen

A source-linked companion for pathology-report wording and care-team conversations

Keep the wording. Make the next conversation clearer.

Lumen explores a narrow, evidence-first way to help people locate report wording and prepare neutral questions - without replacing a clinician.



SYNTHETIC ONLY • ENGLISH ONLY • MODEL OFF

8	46	34 / 34	64 + 10
FICTIONAL REPORTS	**EXPECTED FACTS**	**GOLD SPANS**	**TESTS PASSED**
In the frozen test set	Hand-authored labels	Exact fixture match	Backend + browser

WHAT LUMEN IS

A text-only, English prototype designed to make pathology-report wording easier to discuss. The intended experience centers on exact source excerpts, missing facts marked **not stated**, and neutral questions for a care team.

Current demo: model provider disabled. The live analysis path returns no claims; the positive result view is a synthetic UI test fixture.

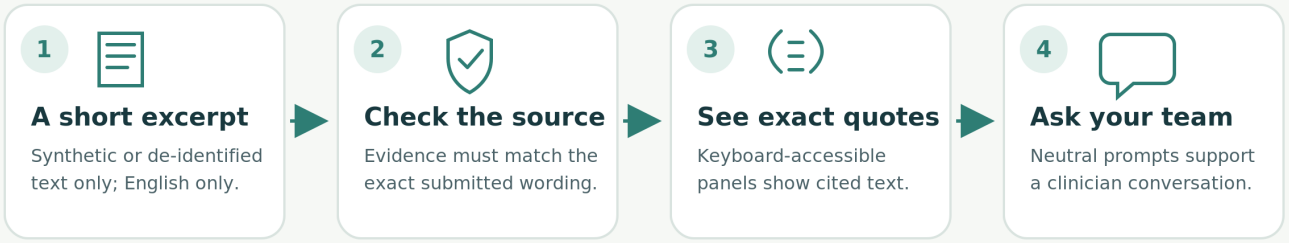
WHAT IT IS NOT

Not a diagnosis, prognosis, treatment recommendation, dose guide, trial-eligibility tool, or urgent-symptom triage service. No clinical outcomes or patient benefit have been measured.

A source-first experience

The product is designed around one simple promise: keep the source visible and support a better conversation, not a care decision.

THE INTENDED EXPERIENCE



SOURCE WORDING → CHECKED EVIDENCE → NEUTRAL DISCUSSION

DESIGNED EXPERIENCE

1. Submit an excerpt
Text-only; explicit synthetic/de-identification and processing acknowledgements.

2. Keep evidence close
Each displayed claim has keyboard-accessible source controls and a labeled exact-quote panel.

3. Prepare a question
Neutral care-team questions keep interpretation with the clinician.

SAFETY BOUNDARIES

The prototype does not infer missing facts: absent fields remain **not stated**. Unsupported or malformed outputs fail closed. Fixed boundaries cover diagnosis beyond the report, prognosis, treatment, dosing, trial eligibility and urgent triage.

English only. No upload, PDF, photo, OCR, translation or external registry integration is offered.

DEMO TRANSPARENCY

The production demo keeps model processing disabled. Submitting through the live browser-to-service path returns a generic unavailable response and no claims. The positive source-panel screen is shown only in a deterministic Playwright UI fixture; it is not AI output, a clinical answer, or an evaluation result.

Trust is a design constraint

Evidence controls, clear refusal boundaries and honest data-flow disclosures are part of the experience - not a claim of clinical safety.

HIGH-LEVEL INFORMATION FLOW

A design view, not a deployment certification



Current demo: model provider disabled

The live path returns a generic unavailable response and no claims. The positive result screen is a synthetic UI test fixture.
If a provider is enabled later, submitted text may leave the device; provider retention/region terms require separate review.

SAFETY LAYERS

Source checking - quote text and offsets are checked against the submitted excerpt.

Constrained responses - structured fields restrict what can be displayed.

Independent verification - a separate verifier stage is designed to approve supported facts.

Fail closed - refusals, no-claim states and generic service errors avoid showing unsupported claims.

PRIVACY BOUNDARY

The prototype has no report database and does not put report text in the URL or browser storage. Checks use fictional text only; application paths are designed not to log report bodies.

Provider status now: disabled; no external model request was made.

If enabled later: submitted text may be sent to an operator-selected service. Provider retention, region and terms have not been reviewed for a deployment.

WHAT THIS DOES NOT PROVE

Passing software tests does not establish model accuracy, clinical correctness, immunity to prompt injection, privacy compliance, regulatory clearance or improved patient outcomes. This prototype is not for clinical decisions or patient use.

Evidence, achievements and next steps

Measured counts below describe software tests and synthetic fixture integrity only. No model-quality or clinical-outcome metric is available.

WHAT HAS BEEN VERIFIED

8	46	34 / 34	64 + 10
fictional reports	**expected field labels**	**gold span matches**	**backend + browser tests**
synthetic corpus v1.0.0	hand-authored answer keys	fixture integrity only	software behavior, not outcomes

NO LIVE MODEL RUN • NO CLINICAL VALIDATION • NO PATIENT OUTCOME CLAIM

VERIFIED IN THIS BUILD

64 backend regression tests passed.
10 browser E2E tests passed, including a provider-disabled real proxy path.
0 axe WCAG 2.1 A/AA violations in the tested form/result states.
34/34 hand-authored gold quote/offset spans match the synthetic source fixtures.

Dependency audits passed after a test-dependency advisory was fixed.

NOT MEASURED

Model extraction precision/recall, unsupported-claim rate before/after verification, model refusal rates, provider latency, translation quality, clinical correctness, patient understanding and health outcomes.

Why: no live model call, no human claim labels, and no clinician review. Treat every model-quality metric as **not measured**.

RESPONSIBLE NEXT STEPS

Potential value (unmeasured): keep source wording visible and help prepare neutral questions. User benefit and clinical impact are unknown.

01 **Define intended use**
with clinicians and patients

02 **Review outputs**
human-labeled synthetic evaluation

03 **Test accessibility**
screen reader and actual zoom

04 **Assess operations**
privacy, deployment and abuse controls

STATUS: SYNTHETIC-ONLY PROTOTYPE • GATE 2 NOT PASSED • NOT FOR PATIENT USE